

Linux Wireless & Network Analysis Reference

Essential Commands & Workflow Cheat Sheet: ip, iw, nmcli, aircrack-ng & Wireshark

1. Network Interface & IP Management (ip & ss)

Command	Description / Purpose
<code>ip a</code> or <code>ip addr</code>	Display all network interfaces and assigned IP addresses (v4 & v6).
<code>ip link show</code>	List status (UP/DOWN), MAC address, and state of all interfaces.
<code>ip route</code>	Display current routing table and default gateway IP address.
<code>sudo ip link set wlo1 down</code>	Bring down the Wi-Fi interface (required before changing modes).
<code>sudo ip link set wlo1 up</code>	Bring up the Wi-Fi interface after configuration changes.
<code>ss -tulpn</code>	List all listening TCP/UDP ports and associated active processes.

2. Wi-Fi Configuration & Monitor Mode (iw)

Command	Description / Purpose
<code>iw dev</code>	Display physical radio devices (<code>phy#</code>) and interface names (<code>wlo1</code>).
<code>iw dev wlo1 info</code>	Show current operating mode, channel, bandwidth, and TX power.
<code>iw dev wlo1 link</code>	Show active AP connection details (SSID, BSSID, Signal dBm, Freq).
<code>sudo iw dev wlo1 scan grep -E 'SSID DS'</code>	Perform over-the-air scan for nearby AP SSIDs and channel numbers.
<code>sudo iw dev wlo1 set type monitor</code>	Set interface mode to Monitor Mode (passive packet capture).
<code>sudo iw dev wlo1 set type managed</code>	Revert interface back to Managed Mode (standard internet Wi-Fi).
<code>sudo iw dev wlo1 set channel 6</code>	Tune adapter to Channel 6 (2.4 GHz, 20 MHz width default).
<code>sudo iw dev wlo1 set channel 149 80MHz</code>	Tune adapter to Channel 149 with an 80 MHz bonded channel width .
<code>sudo iw dev wlo1 set freq 5745 80 5775</code>	Explicitly set control frequency (5745MHz), width (80MHz), and center (5775MHz).
<code>sudo iw dev wlo1 set power_save off</code>	Disable Wi-Fi power saving (prevents interface drops during capture).

💡 5 GHz 80 MHz Bonded Bandwidth Mapping Table

Channels Covered	Control Freq	Width	Center Freq	Exact Command
36, 40, 44, 48	5180 MHz (Ch 36)	80 MHz	5210 MHz	<code>sudo iw dev wlo1 set freq 5180 80 5210</code>
52, 56, 60, 64	5260 MHz (Ch 52)	80 MHz	5290 MHz	<code>sudo iw dev wlo1 set freq 5260 80 5290</code>
149, 153, 157, 161	5745 MHz (Ch 149)	80 MHz	5775 MHz	<code>sudo iw dev wlo1 set freq 5745 80 5775</code>

3. NetworkManager Control (nmcli & systemctl)

Command	Description / Purpose
<code>nmcli device status</code>	Show management status of all adapters (Managed / Unmanaged / Connected).
<code>nmcli device wifi list</code>	Display a clean table of visible Wi-Fi networks, signal strength, and security.
<code>nmcli dev wifi connect "SSID" password "KEY"</code>	Connect to a specific Wi-Fi network from CLI.
<code>sudo nmcli device set wlo1 managed no</code>	Tell NetworkManager to ignore <code>wlo1</code> (prevents interference in monitor mode).
<code>sudo nmcli device set wlo1 managed yes</code>	Restore NetworkManager control over <code>wlo1</code> .
<code>sudo systemctl stop NetworkManager</code>	Stop the NetworkManager service completely for audit sessions.
<code>sudo systemctl start NetworkManager</code>	Restart NetworkManager service for normal internet connection.

4. Wireless Auditing & Packet Injection (aircrack-ng Suite)

Command	Description / Purpose
<code>sudo airmon-ng check kill</code>	Kill interfering background processes (<code>wpa_supplicant</code> , <code>dhclient</code>).
<code>sudo airodump-ng wlo1</code>	Start real-time live scanner showing all nearby APs, channels, and connected clients.
<code>sudo airodump-ng -c 149 --bssid <MAC> -w cap_out wlo1</code>	Lock onto channel 149, filter specific AP BSSID, and record frames/handshakes to file.
<code>sudo aireplay-ng --test wlo1 or -9</code>	Injection Test: Send test probe requests to test hardware & driver packet injection.
<code>sudo aireplay-ng -0 5 -a <AP_MAC> -c <CLIENT_MAC> wlo1</code>	Send 5 deauth frames to force client reconnection (used for handshake capture).

⚠ Interpreting Packet Injection Test Results

- **Success (Injection Working!):** Adapter transmitted raw frames and received replies.
- **Enterprise AP Note (0/30 - 0%):** Enterprise APs (Cisco/Aruba at campuses) run WIDS/WIPS and ignore directed probe requests. Test against a mobile hotspot (2.4 GHz) to verify 100% injection.
- **Chipset Compatibility:** MediaTek (MT7921) & Realtek support injection natively via `mt76` / open drivers. Intel (`iwlwifi`) supports monitor mode but blocks active packet injection.

5. Hardware Inspection, Permissions & Wireshark Configuration

Command	Description / Purpose
<code>lspci -nnk grep -iA3 network</code>	Identify PCI Wi-Fi card model (e.g. MediaTek MT7921) and active kernel driver module.
<code>lsusb</code>	Display attached USB Wi-Fi dongles and chipsets (e.g., RTL8812AU).
<code>groups</code>	Check active user groups (verify inclusion in <code>wheel</code> and <code>wireshark</code>).
<code>sudo usermod -aG wireshark \$USER</code>	Add user to <code>wireshark</code> group (Must use -a flag to prevent losing <code>wheel</code> !).
<code>newgrp wireshark</code>	Instantly apply group permissions in current terminal session without logging out.
<code>sudo setcap cap_net_raw,cap_net_admin=eip /usr/bin/dumpcap</code>	Grant Wireshark capability to capture raw frames and switch interface modes without root.

Command	Description / Purpose
<code>df -h /var/tmp</code>	Verify disk space in temporary directory (where Wireshark writes pcap files).
<code>python3 -m http.server 8000</code>	Launch quick local test web server on port 8000 to generate test HTTP traffic.

6. Wireshark Interface & Display Filter Quick Reference

Scenario / Objective	Interface Choice	Display Filter / Action
Local python server / localhost	<code>Loopback: lo</code>	<code>tcp.port == 8000</code> or <code>http</code>
Wi-Fi Internet traffic	<code>wlo1</code> (Managed Mode)	<code>ip.addr == 10.x.x.x</code> or <code>dns http</code>
Over-the-air raw 802.11 frames	<code>wlo1</code> (Monitor Mode)	<code>wlan</code> or <code>wlan.fc.type_subtype == 0x08</code> (Beacons)
Filter target client MAC	<code>wlo1</code> (Monitor Mode)	<code>wlan.addr == xx:xx:xx:xx:xx:xx</code>
EAPOL Handshake capture	<code>wlo1</code> (Monitor Mode)	<code>eapol</code>

 **Complete Monitor Mode Workflow Command Sequence**

```
# 1. Stop background manager & kill interfering processes
sudo systemctl stop NetworkManager
sudo airmon-ng check kill

# 2. Put adapter into monitor mode & tune channel width
sudo ip link set wlo1 down
sudo iw dev wlo1 set type monitor
sudo iw dev wlo1 set channel 149 80MHz
sudo ip link set wlo1 up

# 3. Verify configuration & start Wireshark
iw dev wlo1 info
wireshark &
```